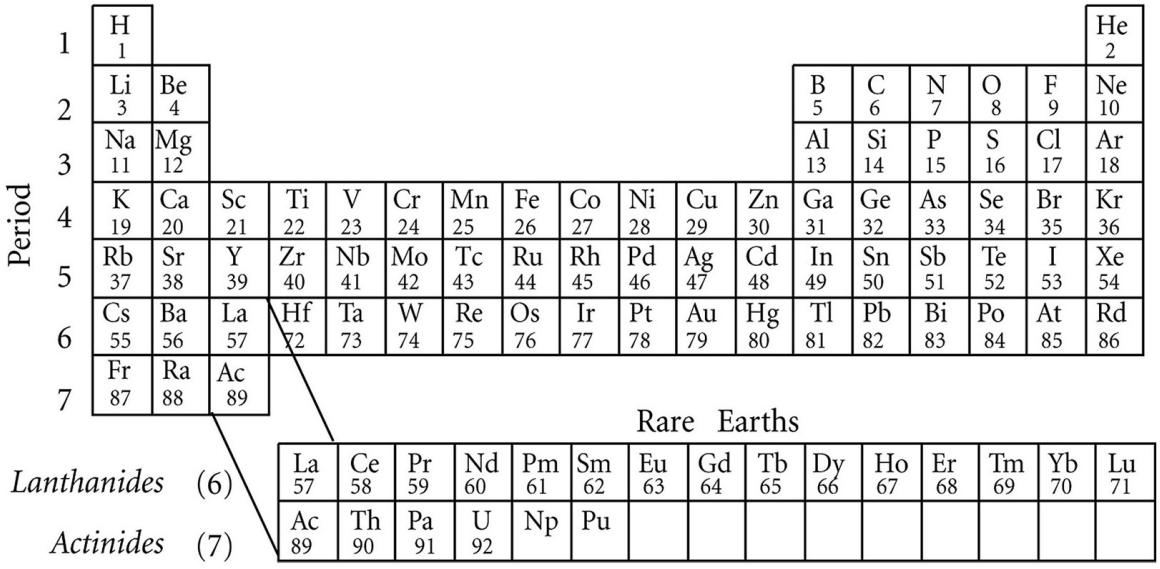
TOOLKIT: Chemistry of the Earth
Practical Two: A Geochemistry view of the Periodic Table

FEEDBACK

Objectives:

- Familiarize yourself with the Periodic Table, its groups and periods
- Review of atomic number, atomic mass, and isotopes
- Understand electron configurations and how they relate to the structure of the Periodic Table
- Practice writing mineral formulae based on charge balance

A. Familiarize yourself with the Periodic Table, its groups and periods



1. Which element am I? (Write the full name of the element and its chemical symbol.)

a) I am an element in the 1st Period. I am a noble gas. Helium, He

b) I am an element in the 2nd Period. I have 4 electrons in my outer shell (two electrons in my 2s orbital and two in my 2p orbital) and I like to form covalent bonds. Carbon, C

c) I am an element in the 2nd Period. I like to form anions with a charge of 2-. Oxygen, O

d) I am an element in the 3rd Period. I like to form cations with a charge of 2+. Magnesium, Mg

e) I am an element in the 3rd Period. I form tetrahedral units in rock-forming minerals. Silicon, Si

f) I am an element in the 4th Period. My 3d orbitals are almost (but not completely) full and I occur as a native metal. Copper, Cu

g) I am an element in the 4th Period. I am a halogen. Bromine, Br

h) I am an element in the 5th Period. I have one electron in my 5s orbital. Rubidium, Rb

i) I am an element in the 6th Period. I have 7 electrons in my 4f orbitals. Europium, Eu

2. Write down at least two questions of your own, along similar lines:

B. Review of atomic number, atomic mass, and isotopes

1. Magnesium (Mg): Z = 12

Magnesium has three isotopes, ^{24}Mg, ^{25}Mg and ^{26}Mg. The relative abundances of the isotopes are given in the Table below:

Mg Isotopes

Isotope	Number of Neutrons (N)	Abundance (%)
^{24}Mg	12	79
^{25}Mg	13	10
^{26}Mg	14	11

How many protons does a magnesium atom contain? ___12_____

Complete the Table to give the number of neutrons in each isotope. Then calculate the atomic mass of magnesium in nature.

Atomic mass of Mg = (24 * 0.79) + (25 * 0.1) + (26 * 0.11) = 24.32

2. Lead (Pb): Z = 82.

Lead has four isotopes, ^{204}Pb, ^{206}Pb, ^{207}Pb, ^{208}Pb. The relative abundances of the isotopes are given in the Table below:

Pb Isotopes

Isotope	Number of Neutrons (N)	Abundance (%)
^{204}Pb	122	1.4
^{206}Pb	124	24.1
^{207}Pb	125	22.1
^{208}Pb	126	52.4

How many protons does a lead atom contain? ___82_____

Complete the Table to give the number of neutrons in each isotope, and the abundance of ^{208}Pb. Then calculate the atomic mass of lead in nature.

Atomic mass of Pb = (204 * 0.014) + (206 * 0.241) + (207 * 0.221) + (208 * 0.524) = 207.241

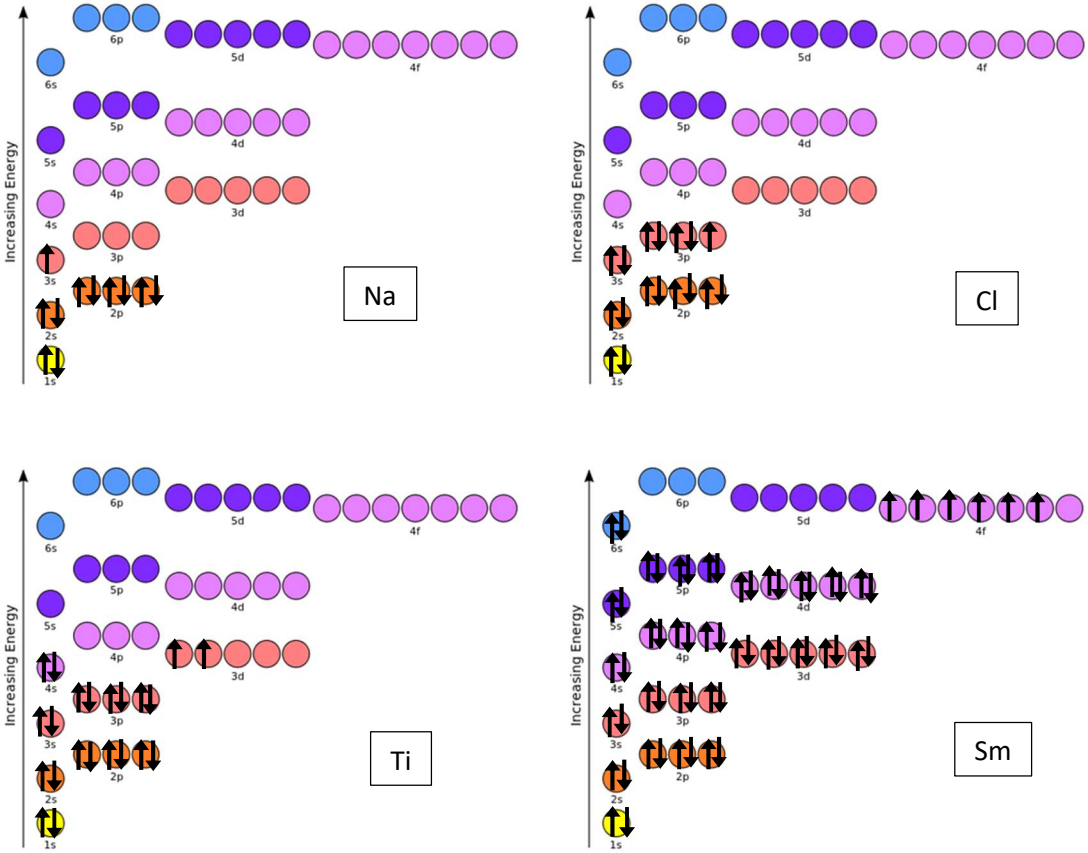
Note that the numbers you calculated for Mg and Pb do not match exactly with the values in your Periodic Table: Mg = 24.305 and Pb = 207.2. A ^{24}Mg atom does not have a mass of exactly 24. The mass is slightly less because the energy taken to bind the nucleus together is equivalent to a small mass ($E = mc^2$). The mass of ^{24}Mg is actually 23.985. The exact equation is as follows:

(23.985*0.7899)+(24.986*0.1)+(25.983*0.1101) = 24.30508

C. Understand electron configurations and how they relate to the structure of the Periodic Table

1. The diagrams show electron orbitals as a function of their increasing energy. On these diagrams show the electron configurations for the following elements:

a) Na (Z = 11), b) Cl (Z = 17), c) Ti (Z = 22), d) Sm (Z = 62)



b) For each element, write the electron configuration in the format of Na:

Na: $1s^2\ 2s^2\ 2p^6\ 3s^1$ This can also be written as: [Ne] $3s^1$

Cl: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ [Ne] $3s^2\ 3p^5$

Ti: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$ (or, $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2\ 4s^2$) [Ar] $4s^2\ 3d^2$

Sm: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^6\ 6s^2\ 4f^6$ [Xe] $6s^2\ 4f^6$

(written in order of shells as $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6\ 4d^{10}\ 4f^6\ 5s^2\ 5p^6\ 6s^2$)

c) For Na, Cl and Ti, predict the most stable ion that each atom will form, and explain why.

Na^+ (easily loses its $3s^1$ electron) Cl^- (easily adds an electron to its 3p orbitals)

Ti^{4+} (loses its 4s and 3d electrons – this is not so obvious)

d) Samarium (Sm) is a Lanthanide, or Rare Earth Element (REE). It usually forms a cation with charge 3+. Predict the electron configuration for a Sm^{3+} ion.

Answer is $6s^2 4f^6$ goes to $6s^0 4f^5$ – again, not obvious. Electron configurations for the REE are complicated, but they are usually all 3+ ions.

2. Which (neutral) atoms have the following electron configurations:

a) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ Ca

b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$ Fe

c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$ Br

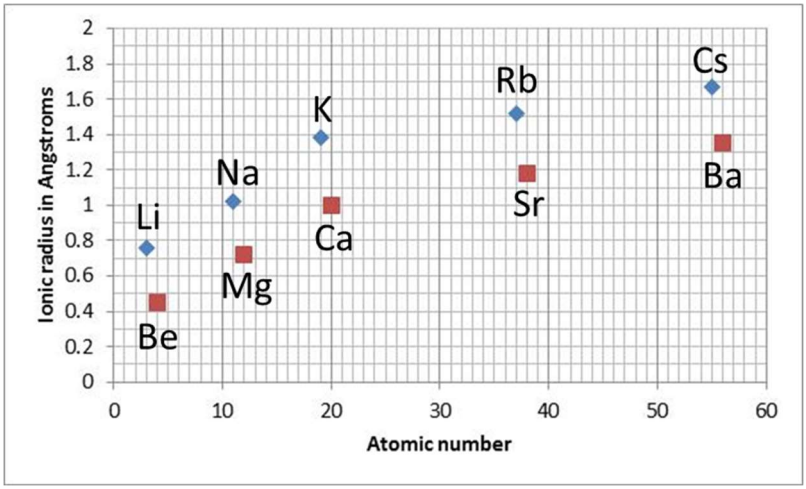
D. Investigate how the properties of elements vary within groups

1. As we go down a group in the Periodic Table, we increase the atomic number of the element. With increasing atomic number, the electron cloud gets larger and the size of the atom increases. In geochemistry, it is important to know the size of an ION. Use the data in the Table below to plot a graph of the change in radius of common ions as we go down Groups 1 and 2 of the Periodic Table.

Atomic number and ionic radius		
	Atomic number	Ionic radius (Angstroms)
GROUP 1		
Li	3	0.76
Na	11	1.02
K	19	1.38
Rb	37	1.52
Cs	55	1.67
GROUP 2		
Be	4	0.45
Mg	12	0.72
Ca	20	1.00
Sr	38	1.18
Ba	56	1.35

2. An Angstrom (Å) is a convenient unit for measuring the size of atoms and ions, but it is not an SI unit. An Angstrom is 10^{-10} metres. How many nanometres is this?

Nano means 10^{-9}, so an Angstrom is 0.1 nm



3. Explain the similarities and differences between Group 1 and Group 2.

The two groups show the same trend of increasing ionic radius with increasing atomic number. The data are for monovalent (1+) ions for Group 1, e.g. Na^+, and divalent (2+) ions for Group 2, e.g. Mg^{2+}. The extra charge in Group 2, which is centred on the nucleus, pulls the electron cloud closer to the nucleus, thus reducing the ionic radius.

4. Which Group 2 ion has the same electron configuration as Na^+? What additional element forms a trivalent (3+) ion with the same electron configuration? How would the size of that ion compare with the size of the Na^+ ion?

Mg^{2+} and Al^{3+} both have the same electron configuration as Na^+ ($1s^2\ 2s^2\ 2p^6$). With the additional net charge on the nucleus (3+), the Al^{3+} ion contracts even more, and is much smaller than Na^+ and Mg^{2+}.

E. Relate the periodic table to mineral formulae

1. The minerals below can all be considered as ionic compounds. For each mineral, write the chemical formula. Determine the charge on each ion based on your knowledge of the periodic table, and balance the formula so that the overall charge is zero.

a) **Halite** – sodium and chlorine

Charge on Na = 1+ Charge on Cl = 1- Formula: NaCl

b) **Fluorite** – calcium and fluorine

Charge on Ca = 2+ Charge on F = 1- Formula: CaF_2

c) **Sylvite** – potassium and chlorine

K^+ Cl^- : KCl

d) **Oldhamite** – calcium and sulphur

Ca^{2+} S^{2-} : CaS

e) **Periclase** – magnesium and oxygen

Mg^{2+} O^{2-} : MgO

f) **Rutile** – titanium and oxygen

Ti^{4+} O^{2-} : TiO_2

g) **Corundum** – aluminium and oxygen

Al^{3+} O^{2-} : Al_2O_3

h) **Spinel** – magnesium, aluminium and oxygen
(Hint: The total cation charge is 8)

Mg^{2+} Al^{3+} O^{2-} : $MgAl_2O_4$

i) **Forsterite –** magnesium silicate (silicate in this instance = $[SiO_4]^{4-}$)

Mg^{2+} SiO_4^{4-} : Mg_2SiO_4

j) Calcite – calcium carbonate (carbonate = $[CO_3]^{2-}$)

Ca^{2+} CO_3^{2-} : $CaCO_3$

k) Barite – barium sulphate (sulphate = $[SO_4]^{2-}$)

Ba^{2+} SO_4^{2-} : $BaSO_4$

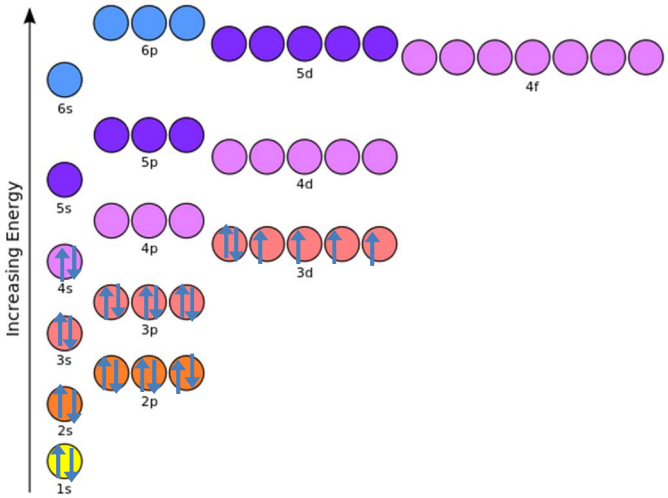
l) **Fluorapatite** – calcium, fluorine and phosphate (phosphate = $[PO_4]^{3-}$)

(Hint: the total cation charge and anion charge is 10)

Ca^{2+} F^- PO_4^{3-} : $Ca_5F(PO_4)_3$ (usually written $Ca_5(PO_4)_3F$)

Note that other combinations would be charge-balanced, e.g. $Ca_5(PO_4)_2F_4$, but the mineral fluorapatite has the specific balance of ions given.

2. **Transition elements.** It is hard to predict the charge on transition elements based on their electron configurations. As an example, let us consider iron (Fe), Z = 26. Complete the electron configuration diagram for iron (the neutral atom) in the diagram below and write the electron configuration as $1s^2 2s^2$.....



Fe: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

Iron commonly forms two ions, Fe^{2+} and Fe^{3+}. In the Fe^{2+} ion, the two 4s electrons are lost. In the Fe^{3+} ion, the two 4s electrons and one of the 3d electrons are lost.

For iron, the Fe^{2+} ion, or divalent iron, is referred to as FERROUS

the Fe^{3+} ion, or trivalent iron, is referred to as FERRIC

The different ions have *oxidation states* of 2 and 3 respectively.

Fe^{2+}: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^6$ Fe^{3+}: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

In the transition series, the 4s electrons are lost most easily, so 2+ ions are common. But all the transition elements can occur with different ion charges. We will not concern ourselves with the details of electron configurations for the transition elements. For the minerals below, assume the following ion charges which are common in minerals:

Ti^{4+}, Cr^{3+}, Mn^{4+}, Fe^{2+} or Fe^{3+}

For each mineral below, write the chemical formula:

a) **Chromite** – iron (Fe^{2+}) chromium oxide

$FeCr_2O_4$

b) **Hematite** – iron (Fe^{3+}) oxide (written as Fe(III) oxide)

Fe_2O_3

c) **Magnetite** – iron oxide - a mixture of Fe^{2+} and Fe^{3+}

$Fe^{2+}Fe_2^{3+}O_4^{2-}$ or Fe_3O_4

d) **Ilmenite** – iron (Fe^{2+}) titanium oxide

$FeTiO_3$

e) **Pyrolusite** – manganese oxide

MnO_2

f) **Siderite** – iron (Fe^{2+}) carbonate

$FeCO_3$

g) **Fayalite** – iron (Fe^{2+}) silicate (silicate in this instance = $[SiO_4]^{4-}$)

Fe_2SiO_4

h) For the following manganese minerals, determine the charges on the transition elements from charge balance considerations:

Rhodonite – $MnCO_3$

$Mn^{2+}CO_3^{2-}$

Pyrophanite – $MnTiO_3$

$Mn^{2+}Ti^{4+}O_3^{2-}$

Alabandite – MnS

$Mn^{2+}S^{2-}$

Hausmannite – Mn_3O_4

$Mn^{2+}Mn^{3+}{}_2O_4$ – compare with magnetite, chromite and spinel above

Hollandite – $BaMn^x{}_6Mn^y{}_2O_{16}$

$BaMn^{4+}{}_6Mn^{3+}{}_2O_{16}$

Spessartine – $Mn_3Al_2(SiO_4)_3$

$Mn_3{}^{2+}Al_2{}^{3+}(SiO_4)_3{}^{4-}$

Tantalite – $(Fe,Mn)Ta_2O_6$ and **Columbite –** $(Fe,Mn)Nb_2O_6$

(Ta and Nb are 5+ in these minerals. Check how this makes sense in the periodic table. Tantalite and columbite are the primary ores of Ta and Nb.)

$(Fe^{2+},Mn^{2+})Ta_2O_6$ $(Fe^{2+},Mn^{2+})Nb_2O_6$

The notation (Fe,Mn) means that Fe and Mn atoms could be present in any proportions in this mineral. Both have the same charge (2+). In any piece of tantalite (or columbite), the total number of (Fe+Mn) atoms compared to Ta (or Nb) atoms is 1:2, and the ratio of Ta (or Nb) to oxygen is 2:6.

i) What factors might influence the formation of manganese minerals (and minerals of other transition elements) with different charges on the ions?

One of the main factors is the amount of oxygen available. We will talk more about "oxidation state" in a later lecture. If more oxygen is available, a transition element ion will have a higher charge, e.g. when there is plenty of oxygen available, iron will tend to form Fe^{3+} rather than Fe^{2+}.The amount of oxygen available also influences things like the state of sulphur (e.g. more oxygen results in sulphate ions, $SO_4{}^{2-}$, rather than sulphide, S^{2-}). The charge on a transition element ion also depends on the mineral in which it is substituting. For example, where Mn is substituting for Fe and Mg in a silicate mineral it will probably be Mn^{2+}, with the same charge as the main elements in the mineral (Fe^{2+} and Mg^{2+}). The size of an ion also has an important effect, for example Mn^{3+} would not easily substitute for Al^{3+} because it's much bigger.